

A New Adaptive Grid Generation by Elliptic Equations with Orthogonality at All of the Boundaries

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A new adaptive grid generation procedure is proposed, which combines a previously developed method that used Anderson's adaptive grid scheme along with a method for controlling grid spacing and orthogonality on all of the boundaries. The proposed method assigns the desired grid stretching over the smooth region during initial grid system generation and before grid adaptation is performed. After properly interpreting the smoothing term of the weighting function, the desired grid stretching is added to the adaptive grid scheme. Several test cases illustrate the method's feasibility.

KEY WORDS: Adaptive grid; elliptic grid generation; orthogonal mesh.

1. INTRODUCTION

Both the finite-difference and finite-volume methods have become increasingly important in the solving of partial differential equations. These numerical methods rely on discretization of the domain of definition, most frequently employing numerical grid generation (for comprehensive reviews of numerical grid generation, the reader is referred to Thompson, Warsi, and Mastin, 1982; Thompson, 1984, 1985; Eiseman, 1987; Hawken, Gottlieb, and Hansen, 1991).

In order to accurately solve a problem numerically, both the proper location of the nodal points of the computational domain and the orthogonal grid around the boundaries are of prime importance. While the adaptive grid method resolves physical variables by moving grid points from areas of small solution variation to regions of larger variation, it does

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not address problems associated with the physical quantity gradient normal to the boundaries. This latter factor is particularly important for friction factor or heat transfer coefficient evaluation. If orthogonality at the mesh points adjacent to the boundaries can be guaranteed, then the error induced from the treatment of the boundary conditions can be decreased.

One of the most popular methods for grid generation involves solving the elliptical partial differential equation; however, the choice of control functions poses a major problem. In 1974, Thompson *et al.* (1974, 1977) tried to overcome this problem by employing an exponential form of the control functions, but, unfortunately, there is as yet no automatic process for selecting their values in a desired clustering. In 1979, Steger and Sorenson (Steger and Sorenson, 1979; Sorenson, 1980) effectively controlled both grid size and intersecting angle, but only at the boundaries $\eta = 0$ and $\eta = \eta_{\max}$. In 1980, Thomas and Middlecoff proposed a direct control method whereby the resulting interior grid distribution is controlled entirely by a prior selection of the grid distribution along all the boundaries. Unfortunately, the angle of intersection between the transverse grid line and the boundaries cannot be controlled as effectively as with Steger and Sorenson's method (Steger and Sorenson, 1979; Sorenson, 1980). By virtue of the Anderson and Steinbrenner curve-by-curve adaptive grid theory (Anderson and Steinbrenner, 1986; Anderson, 1987a), it can be proved that Thomas and Middlecoff's formulation (for interior points near boundaries) does not completely reflect the curvature effects of the boundary.

Sorenson, improving upon his previous work (Steger and Sorenson, 1979; Sorenson, 1980), developed a method in which all the boundaries preserved orthogonality (Sorenson, 1986). His modification was, however, incomplete since the control functions were not expressed as interpolated forms, and consequently, the modification did not always provide the desired grid control at regions corresponding to corners on the computational domain (Hsu and Lee, 1991; Jeng and Liou, 1992b). Hsu and Lee (1991) modified Steger and Sorenson's method, using power law interpolation and a Neumann boundary condition that allows grid points to be moved along the curved boundaries $\xi = 0$ and $\xi = \xi_{\max}$ until orthogonal grids are obtained. The present authors (Jeng and Liou, 1992b) combined the techniques of Sorenson (1986) and Hsu and Lee, (1991) to produce a grid generation system that gives the desired grid control over all the boundaries.

In 1986, Anderson (Anderson and Steinbrenner, 1986; Anderson, 1987a, b) related the control functions to weighting functions in the elliptical partial differential equation and developed his well-known adaptive grid method. Owing to the smoothness of this elliptical-type grid generation

method, the excessive grid distortion inherent in such equidistribution methods as those of Dwyer and co-workers (Dwyer *et al.*, 1980; Dwyer, 1984) can be avoided. However, orthogonality of the boundaries is not considered. Recently, the present authors (Jeng and Liou, 1992a) combined the techniques of Jeng and Liou (1992b) and Anderson (1987b) to produce an adaptive method with orthogonality along all boundaries.

Usually, the distribution of an initial grid is designed in such a way as to stretch grid points in regions of smooth solution and far field, so that the total number of grid points for calculation is kept within a reasonable range. However, the adaptive grid methods of Anderson and Steinbrenner (1986), Anderson (1987a, b), and Jeng and Liou (1992a) are not able to effectively give this desired grid stretching.

The present investigation extends our previous work (Jeng and Liou, 1992a), which combined the adaptive grid procedure of Anderson (Anderson and Steinbrenner, 1986; Anderson, 1987a) with our boundary grid control method (Jeng and Liou, 1992b), proposing a new strategy for the choice of weighting function such that the basic features of the original grid are retained. For the sake of clarity, both the adaptive grid and boundary grid control methods are first restated. Finally, several numerical tests are examined in order to demonstrate the applicability of the proposed method.

2. Theoretical Development

2.1. A Review of the Boundary Grid Control Method

The partial differential equations which define the mapping between an irregular region on the (x, y) plane (physical domain) and a regular region on the (ξ, η) plane (computational domain) (Thompson *et al.*, 1974, 1977, 1982) can be expressed as

$$\begin{aligned}\xi_{xx} + \xi_{yy} &= P \\ \eta_{xx} + \eta_{yy} &= Q\end{aligned}\tag{2.1}$$

where $\xi_{xx} = \partial^2 \xi / \partial x^2, \dots$, and P and Q are the control functions. Without loss of generality, consider the problem of using a rectangular region on the computational domain. By interchanging the dependent and independent variables, Eq. (2.1) can be rewritten (Thomas and Middlecoff, 1980) as

$$\begin{aligned}\alpha(x_{\xi\xi} + \phi x_{\xi}) - 2\beta x_{\xi\eta} + \gamma(x_{\eta\eta} + \psi x_{\eta}) &= 0 \\ \alpha(y_{\xi\xi} + \phi y_{\xi}) - 2\beta y_{\xi\eta} + \gamma(y_{\eta\eta} + \psi y_{\eta}) &= 0\end{aligned}\tag{2.2}$$

where

$$\begin{aligned}\alpha &= x_\eta^2 + y_\eta^2, & \beta &= x_\xi x_\eta + y_\xi y_\eta \\ \gamma &= x_\xi^2 + y_\xi^2, & J &= x_\xi y_\eta - x_\eta y_\xi \\ P &= \phi(\xi, \eta)(\xi_x^2 + \xi_y^2) \\ Q &= \psi(\xi, \eta)(\eta_x^2 + \eta_y^2)\end{aligned}$$

Suppose the grid size $(s_\eta)_0 = (ds/d\eta)|_0 \sim \Delta s/\Delta\eta$ next to the boundary line $\eta=0$, as well as the intersection angle θ_0 between the boundary and the transverse grid line $\xi = \text{const}$ were specified (Steger and Sorenson, 1979; Sorenson, 1980; Hsu and Lee, 1991), respectively, as

$$(s_\eta)_0 = (x_\eta^2 + y_\eta^2)_0^{1/2} \quad (2.3)$$

$$(\nabla\xi \cdot \nabla\eta)_0 = |\nabla\xi| |\nabla\eta| \cos(\pi - \theta_0) \quad (2.4)$$

The first order and second order derivatives, then, are

$$\begin{aligned}x_\eta|_0 &= \left[\frac{s_\eta(x_\xi \cos \theta - y_\xi \sin \theta)}{(x_\xi^2 + y_\xi^2)^{1/2}} \right]_0 \\ y_\eta|_0 &= \left[\frac{s_\eta(y_\xi \cos \theta + x_\xi \sin \theta)}{(x_\xi^2 + y_\xi^2)^{1/2}} \right]_0 \\ x_{\eta\eta}|_0 &= \frac{-7x_0 + 8x_1 - x_2}{2(\Delta\eta)^2} - \frac{3(x_\eta)_0}{\Delta\eta} \\ y_{\eta\eta}|_0 &= \frac{-7y_0 + 8y_1 - y_2}{2(\Delta\eta)^2} - \frac{3(y_\eta)_0}{\Delta\eta}\end{aligned} \quad (2.5)$$

The value of control functions ϕ and ψ at $\eta=0$ can now be solved from Eq. (2.2):

$$\begin{aligned}\phi_0 &= \left[\frac{-R_1 y_\eta + R_2 x_\eta}{\alpha J} \right]_0 \\ \psi_0 &= \left[\frac{R_1 y_\xi - R_2 x_\xi}{\gamma J} \right]_0\end{aligned}$$

where

$$\begin{aligned}R_1 &= [\alpha x_{\xi\xi} - 2\beta x_{\xi\eta} + \gamma x_{\eta\eta}]_0 \\ R_2 &= [\alpha y_{\xi\xi} - 2\beta y_{\xi\eta} + \gamma y_{\eta\eta}]_0\end{aligned} \quad (2.6)$$

At this point, note that solving for x_ξ , y_ξ , $x_{\xi\xi}$, and $y_{\xi\xi}$ at the specified grid points on the boundaries, i.e., $(x_{\xi\eta})_0$, $(y_{\xi\eta})_0$, α_0 , β_0 , γ_0 in Eq. (2.5), will give all the terms on the right-hand side of Eq. (2.6). Thus, the interior functions ϕ and ψ (and hence P and Q) can be determined by the following transfinite interpolations (Jeng and Liou, 1992b):

$$\begin{aligned}
 \phi(\xi, \eta) = & \left[\phi(\xi, 0) \left(1 - \frac{\eta}{\eta_{\max}} \right)^{a_1} + \phi(\xi, \eta_{\max}) \left(\frac{\eta}{\eta_{\max}} \right)^{b_1} \right] \\
 & + \left[\phi(0, \eta) \left(1 - \frac{\xi}{\xi_{\max}} \right)^{c_1} + \phi(\xi_{\max}, \eta) \left(\frac{\xi}{\xi_{\max}} \right)^{d_1} \right] \\
 & - \left\{ \left[\phi(0, 0) \left(1 - \frac{\eta}{\eta_{\max}} \right)^{a_1} + \phi(0, \eta_{\max}) \left(\frac{\eta}{\eta_{\max}} \right)^{b_1} \right] \right. \\
 & \times \left(1 - \frac{\xi}{\xi_{\max}} \right)^{c_1} + \left[\phi(\xi_{\max}, 0) \left(1 - \frac{\eta}{\eta_{\max}} \right)^{a_1} \right. \\
 & \left. \left. + \phi(\xi_{\max}, \eta_{\max}) \left(\frac{\eta}{\eta_{\max}} \right)^{b_1} \right] \left(\frac{\xi}{\xi_{\max}} \right)^{d_1} \right\} \\
 \psi(\xi, \eta) = & \left[\psi(\xi, 0) \left(1 - \frac{\eta}{\eta_{\max}} \right)^{a_2} + \psi(\xi, \eta_{\max}) \left(\frac{\eta}{\eta_{\max}} \right)^{b_2} \right] \\
 & + \left[\psi(0, \eta) \left(1 - \frac{\xi}{\xi_{\max}} \right)^{c_2} + \psi(\xi_{\max}, \eta) \left(\frac{\xi}{\xi_{\max}} \right)^{d_2} \right] \\
 & - \left\{ \left[\psi(0, 0) \left(1 - \frac{\eta}{\eta_{\max}} \right)^{a_2} + \psi(0, \eta_{\max}) \left(\frac{\eta}{\eta_{\max}} \right)^{b_2} \right] \right. \\
 & \times \left(1 - \frac{\xi}{\xi_{\max}} \right)^{c_2} + \left[\psi(\xi_{\max}, 0) \left(1 - \frac{\eta}{\eta_{\max}} \right)^{a_2} \right. \\
 & \left. \left. + \psi(\xi_{\max}, \eta_{\max}) \left(\frac{\eta}{\eta_{\max}} \right)^{b_2} \right] \left(\frac{\xi}{\xi_{\max}} \right)^{d_2} \right\} \quad (2.7)
 \end{aligned}$$

where $\phi(\xi, 0)$, $\phi(\xi, \eta_{\max})$, $\psi(\xi, 0)$, and $\psi(\xi, \eta_{\max})$ are evaluated at the corresponding boundaries for $\eta=0$ and $\eta=\eta_{\max}$, while $\phi(0, \eta)$, $\phi(\xi_{\max}, \eta)$, $\psi(0, \eta)$, and $\psi(\xi_{\max}, \eta)$ are evaluated at the boundaries for $\xi=0$, $\xi=\xi_{\max}$ by calculations similar to those in Eqs. (2.3)–(2.6). Since “ ϕ ” and “ ψ ” are the principal control functions along the $\eta=\eta_j$ and $\xi=\xi_i$ lines, respectively, (Anderson *et al.*, 1984; Visbal and Knight, 1982), setting them to be smooth along their respective boundaries should not disturb the boundary specifications. Consequently, “ ϕ ” and “ ψ ” at the corners can be determined the following relations:

$$\begin{aligned}
\phi(0, 0) &= \phi(0, \Delta\eta), & \phi(\xi_{\max}, 0) &= \phi(\xi_{\max}, \Delta\eta) \\
\phi(0, \eta_{\max}) &= \phi(0, \eta_{\max} - \Delta\eta), & \phi(\xi_{\max}, \eta_{\max}) &= \phi(\xi_{\max}, \eta_{\max} - \Delta\eta) \\
\psi(0, 0) &= \psi(\Delta\xi, 0), & \psi(\xi_{\max}, 0) &= \psi(\xi_{\max} - \Delta\xi, 0) \\
\psi(0, \eta_{\max}) &= \psi(\Delta\xi, \eta_{\max}), & \psi(\xi_{\max}, \eta_{\max}) &= \psi(\xi_{\max} - \Delta\xi, \eta_{\max})
\end{aligned} \tag{2.8}$$

The powers " $a_1, b_1, c_1, d_1, a_2, b_2, c_2, d_2$ " are used to specify the variation in the interior control functions. Note that Eq. (2.7) reduces to the linear transfinite interpolation formula when $a_1 = b_1 = c_1 = d_1 = a_2 = b_2 = c_2 = d_2 = 1$.

2.2. Review of Anderson's Adaptive Grid Method

In 1986, by relating the grid control functions to weighting functions, Anderson and Steinbrenner (Anderson and Steinbrenner, 1986; Anderson, 1987a) devised a method of using Poisson grid generation equations [i.e., Eq. (2.1)] to construct adaptive grids. Their idea came from the one-dimensional equidistribution method, which requires

$$S_\xi W = C_1 \tag{2.9}$$

where S is the arc length along a constant η curve, ξ is the coordinate varying along the curve, W is the weighting function, and C_1 is a constant. Such a condition will make the grid spacing along the curve small where W is large and vice versa. If Eq. (2.2) is separately rewritten along constant ξ and constant η curves, the resulting equations can be related to the one-dimensional equidistribution law of Eq. (2.9). Consequently, the relationships between the weighting and grid control functions are

$$\begin{aligned}
P &= \frac{\alpha}{J^2} \frac{1}{W_1} \frac{\partial W_1}{\partial \xi} \\
Q &= \frac{\gamma}{J^2} \frac{1}{W_2} \frac{\partial W_2}{\partial \eta}
\end{aligned} \tag{2.10}$$

where W_1 and W_2 are the weighting functions controlling the grid spacing along $\eta = \text{const}$ and $\xi = \text{const}$ curves, respectively. Since W_1 and W_2 are defined by the solution of the physical problem, the final adaptive grid distribution will reflect the important features of the physical process.

2.3. Adaptive Grid Method with Grid Control on All of the Boundaries

In order to preserve both geometric control and adaptive characteristics, it is advantageous to use the following form (Nielson and Skovgard, 1988; Luong *et al.*, 1991):

$$\begin{aligned} P &= C_W P_g + (1 - C_W) P_W \\ Q &= C_W Q_g + (1 - C_W) Q_W \end{aligned} \quad (2.11)$$

where P_g , Q_g are geometric control functions, P_W , Q_W are Anderson's control functions based on weighting functions, and C_W (Nielson and Skovgard, 1988; Luong *et al.*, 1991) determines the influence of the adaptive control relative to the geometric control. In order to guarantee the desired boundary control (i.e., $P_{\text{boundary}} = P_g$ and $Q_{\text{boundary}} = Q_g$), the following transfinite interpolation for C_W is proposed:

$$\begin{aligned} C_W &= \left\{ \left[1 - \left(\frac{\xi}{\xi_{\max}} \right)^4 \right]^a + \left[1 - \left(1 - \frac{\xi}{\xi_{\max}} \right)^4 \right]^a \right\} \\ &\quad + \left\{ \left[1 - \left(\frac{\eta}{\eta_{\max}} \right)^4 \right]^b + \left[1 - \left(1 - \frac{\eta}{\eta_{\max}} \right)^4 \right]^b \right\} \\ &\quad - \left\{ \left[1 - \left(\frac{\xi}{\xi_{\max}} \right)^4 \right]^a + \left[1 - \left(1 - \frac{\xi}{\xi_{\max}} \right)^4 \right]^a \right\} \\ &\quad \times \left\{ \left[1 - \left(\frac{\eta}{\eta_{\max}} \right)^4 \right]^b + \left[1 - \left(1 - \frac{\eta}{\eta_{\max}} \right)^4 \right]^b \right\} \\ &\quad a = 0.01 \xi_{\max}^4, \quad b = 0.01 \eta_{\max}^4 \end{aligned} \quad (2.12)$$

where $\Delta\xi = \Delta\eta = 1$. Note that the term $[1 - (\xi/\xi_{\max})^4]^a$ becomes negligible as $\xi \geq 5$. Consequently, geometric control dominates near boundary regions, while away from these areas adaptive control takes over.

In order to have effective adaptive control over the entire domain of definition, the boundary grid distribution cannot be arbitrarily specified. In this study, the one-dimensional equidistribution method of Dwyer and co-workers (Dwyer *et al.*, 1980; Dwyer, 1984) and Jeng and Liou's (1989) smoothing are applied in order to distribute the grid points along all the boundaries and determine the grid spacing next to them.

2.4. The Proposed Weighting Function

The one-dimensional equidistribution method of Dwyer and co-workers (Eiseman, 1987; Dwyer *et al.*, 1980; Dwyer, 1984) assumed weighting of the form

$$W_i = 1 + \sum_{j=1}^m \lambda_j M_{ij} \quad (2.13)$$

where the M_{ij} term represents the measurement of a physical variable's variation, which can either be a gradient or second-order derivative of the variable, m is the number of measurements, λ_j is an adaptive constant, and "1" is the smoothing portion. When the term $\lambda|\cdot|$ is set to zero, this "1" results in grids with a uniform grid spacing along a grid line. Note that when λ is large, the final grid possesses a stronger adaptivity with significant clustering in high gradient regions, while a smaller λ introduces a weaker grid adaptivity, resulting in a more smoothly distributed final grid.

An inspection of Eq. (2.13) reveals that no grid stretching factor is involved. To the authors' experience, adding an extra term cannot simultaneously include grid adaptation, grid smoothness, and all the desired stretching. On the other hand, the original grid generation procedure without adaptivity can properly handle both the smoothness and desired stretching. Therefore, *the present study assigns the desired grid stretching during initial grid system generation*. Thus, Eq. (2.13) can be modified to the following form:

$$W_i = \frac{1}{\Delta S_i} + \sum_{j=1}^m \lambda_j M_{ij} \quad (2.14)$$

where ΔS_i is the grid spacing along an initial grid line. If all λ_j are zero, then the $1/\Delta S_i$ term given in Eq. (2.14) will keep the original grid distribution. For a properly defined λ_j , the resulting adaptive grids will preserve the basic properties of the original grid to a certain extent. Although the smooth term of Eq. (2.13) is modified to be $1/\Delta S_i$ in Eq. (2.14), the effect of λ_i on the adaptive behavior is almost the same as will be discussed in more detail in the next section.

3. RESULTS AND DISCUSSION

The first example concerns inviscid flow in a channel with a circular bump, whose thickness-to-chord ratio was chosen to be 10%, while the height of the channel was set equal to the length of the bump. The inlet Mach number was taken as 0.675, and the outlet considered to be subsonic and supported by suitable back pressure. For this special problem, the most interesting characteristics are located around the bump, the solutions in those regions remote from the bump being relatively smooth.

Generally speaking, for the subsonic and transonic problems, in order to get a numerical result independent of the inlet and outlet boundary conditions, the inlet and outlet boundaries of the computational domain should be taken as far from the region of interest as possible. On the other hand, for the sake of saving computer resource, a smaller number of grid

points is preferred. In order to compromise these two requirements, grid points are often stretched in regions of smooth solution. A typical example is shown in the initial grid distribution (65×17) of Fig. 1, in which the grid points are properly stretched from the bump to either the inlet or outlet boundary. A solution found by a finite volume formulated TVD scheme (Jeng *et al.*, 1990) employing this initial grid system is demonstrated in Fig. 2, where the shock is relatively smeared.

Here, the weighting function is defined as

$$W = \frac{1}{\Delta S} + \lambda \left| \frac{\partial p}{\partial S} \right| \quad (3.1)$$

where p represents pressure and λ is the adaptive parameter. In order to examine the effect of λ , three different values (10, 30, and 60) are considered, their resulting adaptive grids being shown in Figs. 3–5, while their corresponding solutions are illustrated in Figs. 6–8. A comparison of Figs. 3–5 with Fig. 1 reveals that the initial grid stretching and clustering characteristics are preserved, as is grid orthogonality around all of the boundaries. Note that for the adaptive grid, grid distribution near the shock front becomes more and more clustered as λ increases, while grid smoothness at either end of the bump degenerates. The associated shock resolution is correspondingly improved, although the solution in most of the other regions remains relatively unchanged. For λ values greater than 60, the grid becomes excessively skewed at both the bump ends, eventually leading to bad solutions in these areas.

The second example concerns a planar two-dimensional incompressible nonsymmetrical divergent channel flow (Shyy, 1986, 1987), which is solved by the SIMPLER algorithm (Jeng and Chen, 1992) using the second-order upwind scheme. Shyy found that the fine grid solution of this system has a separation bubble on the lower surface (Shyy, 1986, 1987). Again, fluid properties near the outlet boundary are relatively smooth, and thus the initial grid distribution is properly stretched, as shown in Fig. 9 (34×26). However, the grid points are not adapted to the solution, and therefore cannot capture the separation bubble (see Fig. 10). For this problem, the pressure term of Eq. (3.1) is replaced by the velocity term u , while the adaptive parameters λ along the ξ and η directions are different: λ_ξ along the ξ direction takes the value of 4; and along the η direction, $\lambda_\eta = 20$ and 30. The corresponding grid distributions for $\lambda_\eta = 20$ and 30 are shown in Figs. 11 and 12, respectively, where the grid orthogonality around all of the boundaries as well as interior smoothness and preserved grid stretching along the x direction are all obvious. Their associated final solutions are illustrated in Figs. 13 and 14, respectively. Note that while

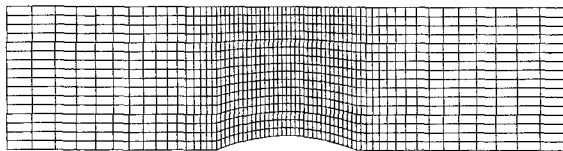


Fig. 1. The initial grid distribution for inviscid flow over a bump, 65×17 , $M_{\text{inlet}} = 0.675$.

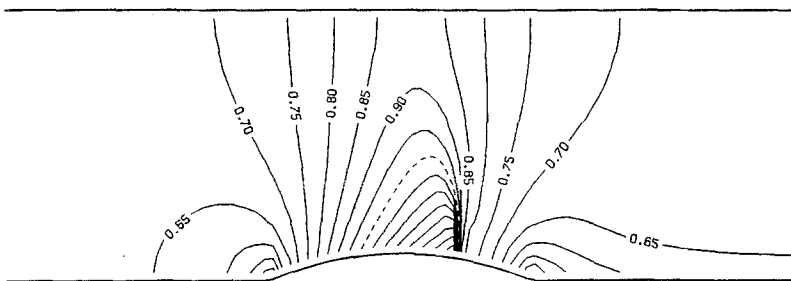


Fig. 2. Mach contour found in Fig. 1 of the initial grid.

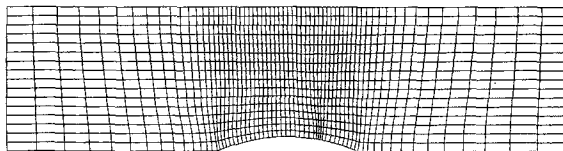


Fig. 3. The adaptive grid generated by the present adaptive grid method with $\lambda = 10$, 65×17 , $M_{\text{inlet}} = 0.675$.

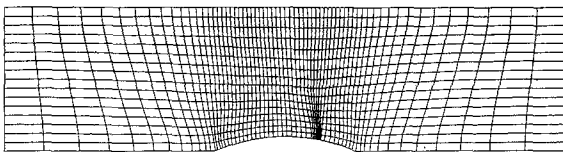


Fig. 4. The adaptive grid generated by the present adaptive grid method with $\lambda = 30$, 65×17 , $M_{\text{inlet}} = 0.675$.

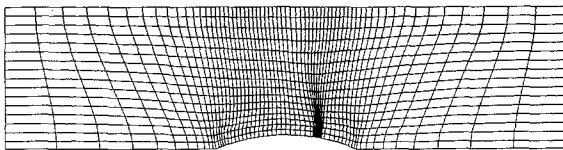


Fig. 5. The adaptive grid generated by the present adaptive grid method with $\lambda = 60$, 65×17 , $M_{\text{inlet}} = 0.675$.

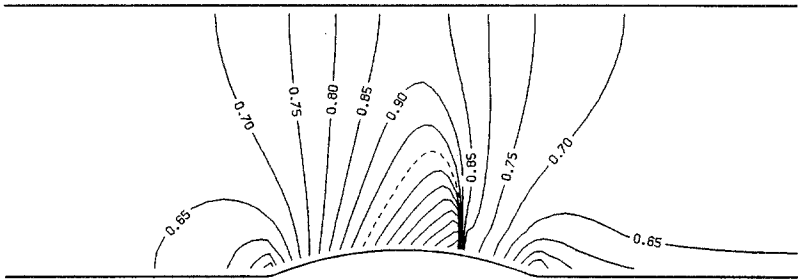


Fig. 6. Mach contour found in Fig. 3 of the present method with $\lambda = 10$.

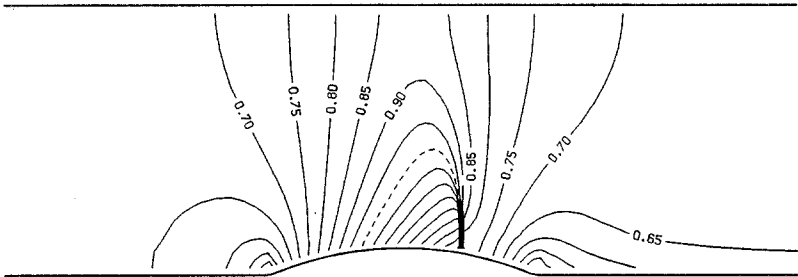


Fig. 7. Mach contour found in Fig. 4 of the present method with $\lambda = 30$.

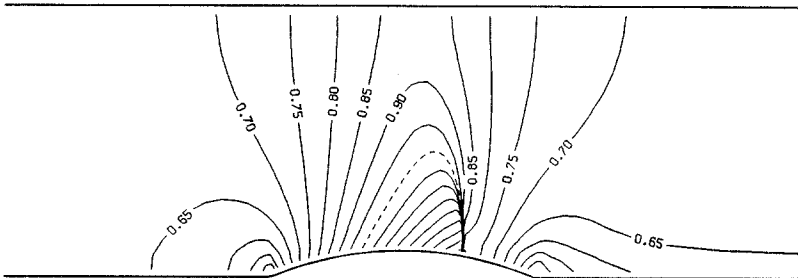


Fig. 8. Mach contour found in Fig. 5 of the present method with $\lambda = 60$.



Fig. 9. The initial grid distribution of divergent channel flow, 34×26 .



Fig. 10. The solution of divergent channel flow obtained from the grid of Fig. 9.



Fig. 11. The adaptive grid of divergent channel flow with $\lambda = 20$, 34×26 .



Fig. 12. The adaptive grid of divergent channel flow with $\lambda = 30$, 34×26 .



Fig. 13. The solution of divergent channel flow obtained from the grid of Fig. 11 with $\lambda = 20$.



Fig. 14. The solution of divergent channel flow obtained from the grid of Fig. 12 with $\lambda = 30$.

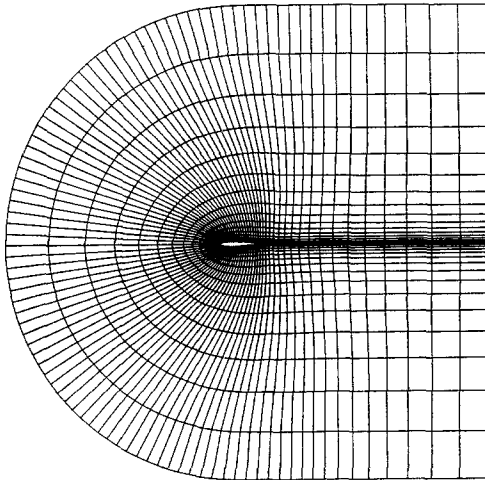


Fig. 15. The initial coarse grid distribution for transonic flow around a NACA 0012 airfoil, 97×17 (overall grid).

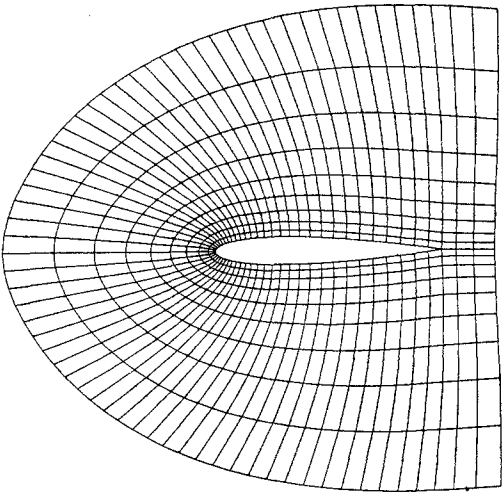


Fig. 16. The initial coarse grid distribution for transonic flow around a NACA 0012 airfoil, 97×17 (nearfield grid).

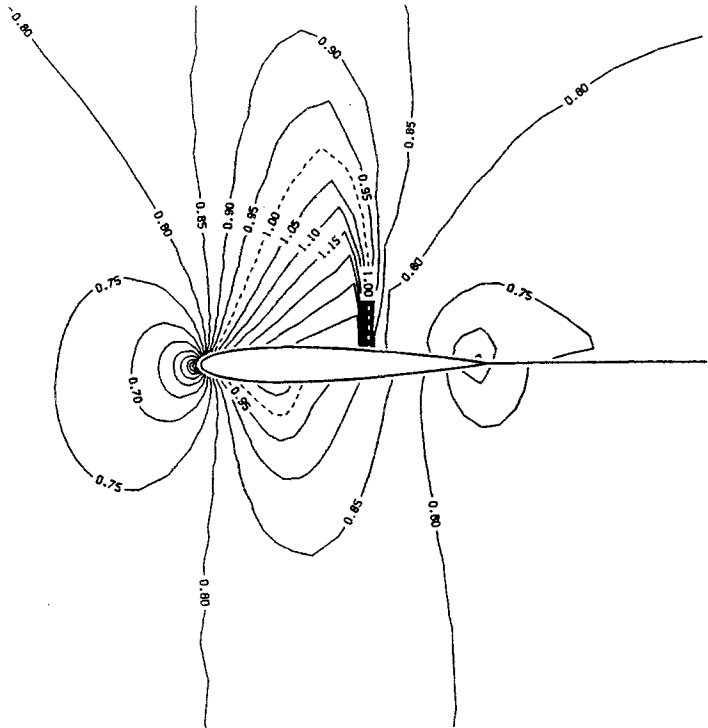


Fig. 17. Mach contour found in Fig. 15 of the initial grid.

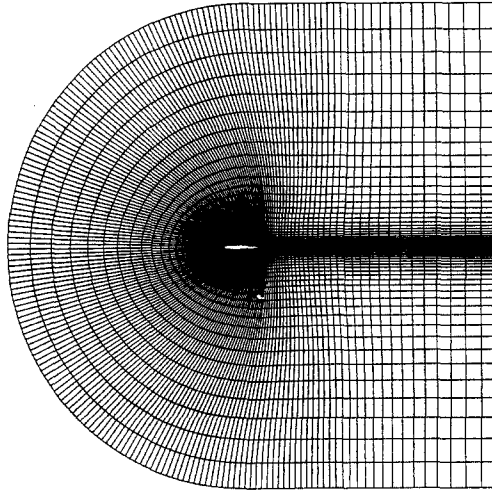


Fig. 18. The fine grid distribution for transonic flow around a NACA 0012 airfoil, 193×33 (overall grid).

they both capture the separation bubble [see Fig. 13 Shyy (1987) for comparison), the former gives a slightly less favorable shorter one than the latter. However, a still larger λ_η will lead to an unacceptable grid distortion, so it seems that $\lambda_\eta = 30$ is the best choice.

The last example considers a steady two-dimensional inviscid transonic flow field around a NACA0012 airfoil with a 0.8 free stream

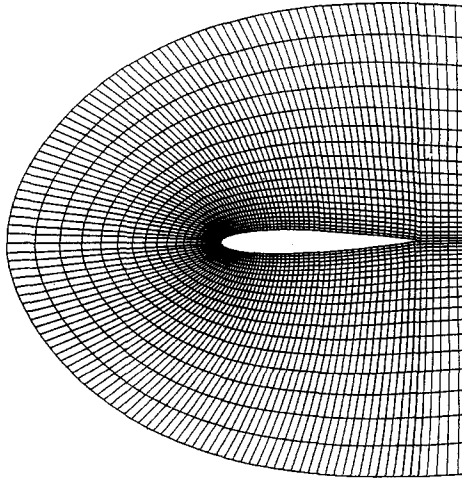


Fig. 19. The fine grid distribution for transonic flow around a NACA 0012 airfoil, 193×33 (nearfield grid).

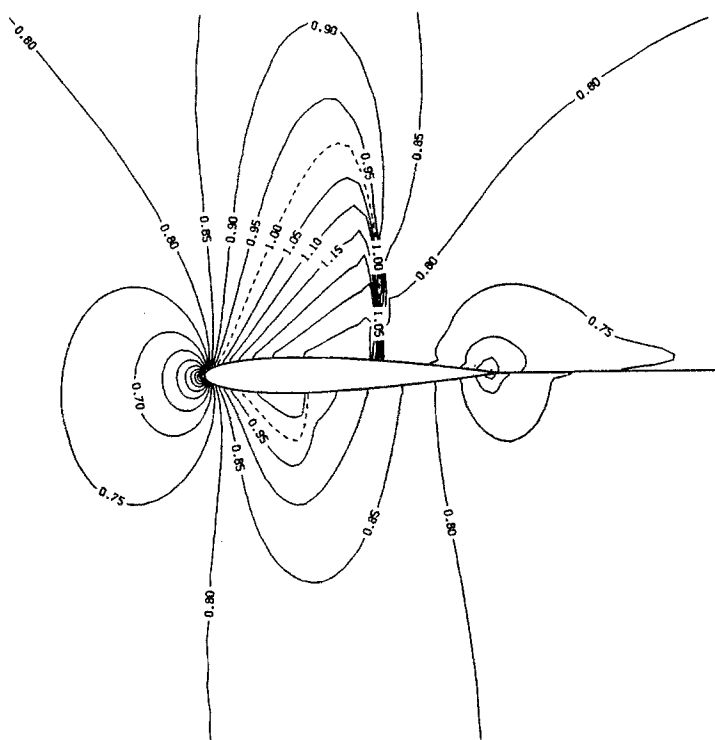


Fig. 20. Mach contour found in Fig. 18 of the fine grid.

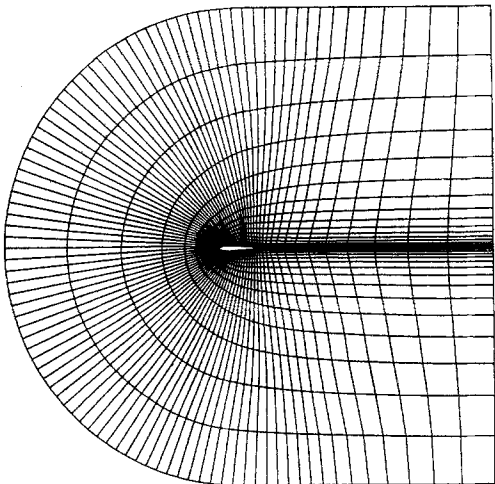


Fig. 21. The adaptive grid of the present method, 97×17 (overall grid).

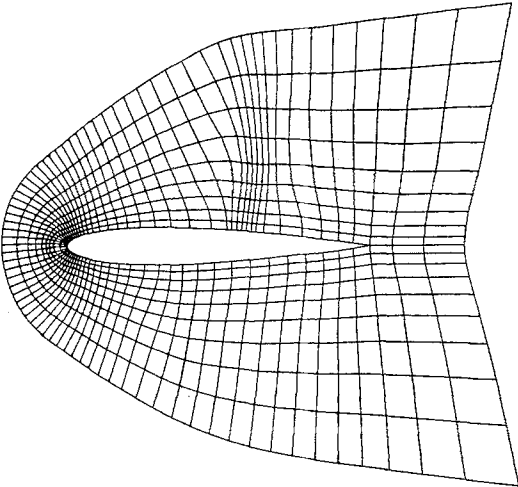


Fig. 22. The adaptive grid of the present method, 97×17 (nearfield grid).

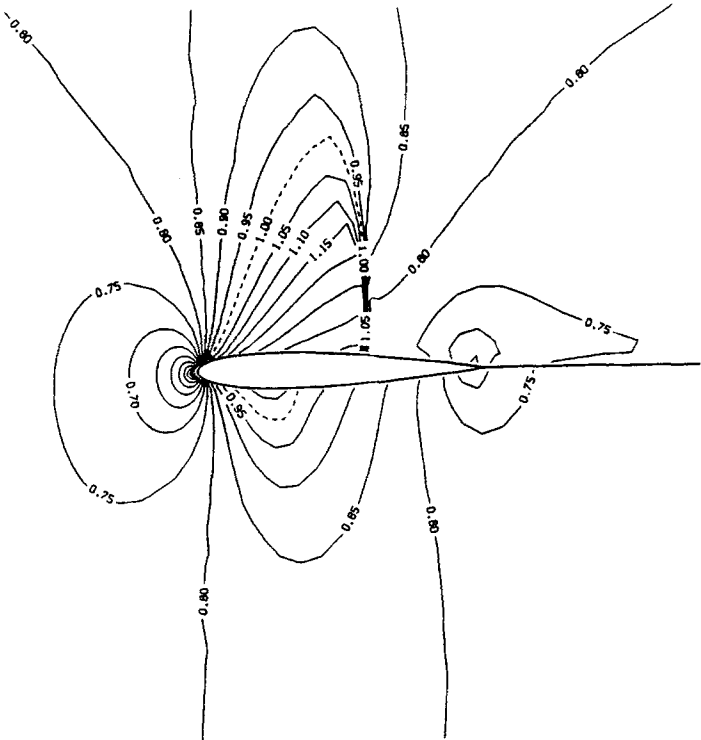


Fig. 23. Mach contour found in Fig. 21 of the present method.

Mach number and an attack angle of 1.25° . The original grid distribution (97×17) is arranged to stretch along both the ξ and η directions as shown in Figs. 15 and 16, while its solution is illustrated in Fig. 17. Such a grid-stretching character is often used because the most interesting part is located around the airfoil surface. As a reference, the fine grid distribution (193×33) and its solution are shown in Figs. 18, 19, and 20, respectively. The adaptive grid system, whose grid distributions are illustrated in Figs. 21 and 22, employs $\lambda = 36$ in Eq. (3.1). The boundary orthogonality and grid-stretching character of the initial grid system are, again, obvious. The final solution shown in Fig. 23 demonstrates that the shock resolution is sharper than that of the fine grid solution, while the resolution in the rest of the region is maintained at a reasonable level.

In all of the test problems, in addition to their adaptive nature, the present numerical scheme suggests both the boundary orthogonality and interior smoothness, while preserving the grid stretching of the initial grid system. It seems that the present study does indeed introduce a new function to the adaptive grid generation method using the elliptic solver.

4. CONCLUSION

A new adaptive grid method was developed, combining a technique for boundary grid control on all of the boundaries, and the Anderson adaptive grid method. The requirement of stretching grid points over regions of smooth solution, which is relatively easily imposed during initial grid generation, is achieved by properly defining the weighting function.

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